VISUAL FRAMEWORK · PRATHAMESH PHUTANE

Machine Learning Algorithm Framework

Twelve widely used algorithms, organized by how they learn and where they are applied: tabular data, computer vision, natural language processing and generative AI.

HOW TO READ THIS FRAMEWORK

- **Supervised:** learns from examples that already have the correct answer (labels)
- **Unsupervised:** finds structure in data without labels (includes self-supervised learning)
- **Primary domain:** where the algorithm is most commonly used
- **Also used:** a real but less common application

01 Algorithms at a Glance

Each row is one algorithm. The dots show which domains it serves, so it is easy to see that tabular data is dominated by classical methods while vision, language and generative AI rely on deep learning.

ALGORITHM	TABULAR DATA	COMPUTER VISION	NLP	GENERATIVE AI	TYPICAL OUTPUT
SUPERVISED LEARNING					
Linear Regression	●				A number
Logistic Regression	●		○		A probability and a class
Decision Tree	●				A class or number, with readable rules
Random Forest	●				A class or number, by majority vote
Gradient Boosting (XGBoost)	●				A class, number or ranking score
K-Nearest Neighbors	●	○			The label of the closest examples
Convolutional Neural Network		●		○	An image label, location or mask
UNSUPERVISED AND SELF-SUPERVISED LEARNING					
K-means Clustering	●	○			Group assignments
Principal Component Analysis	●	○			Fewer, combined features
Transformer (BERT, GPT)		○	●	●	Text understanding or generated text
Generative Adversarial Network		●		●	New realistic images
Diffusion Model		●		●	New images from a text prompt

MOST COMMON CHOICE IN EACH DOMAIN TODAY

TABULAR DATA	COMPUTER VISION	NLP	GENERATIVE AI
Tree ensembles (Random Forest, XGBoost)	**CNNs**	**Transformers**	**Transformers, GANs and Diffusion Models**

02 How Each Algorithm Works

A short explanation of each algorithm, with real-world examples of where it is used today.

SUPERVISED learns from labeled examples

Linear Regression

Supervised | Tabular

HOW IT WORKS

Fits the straight line (or flat plane) that best matches the data by minimizing the squared gap between predicted and actual values.

REAL-WORLD USE

Estimating home prices from size and location; forecasting monthly sales.

Logistic Regression

Supervised | Tabular | NLP

HOW IT WORKS

Adds up weighted features, then passes the total through an S-shaped sigmoid curve to produce a probability between 0 and 1.

REAL-WORLD USE

Predicting whether a borrower will default; filtering spam emails based on word counts.

Decision Tree

Supervised | Tabular

HOW IT WORKS

Asks a series of yes or no questions about the features, choosing each split to make the resulting groups as pure as possible, until it reaches a final answer.

REAL-WORLD USE

Loan approval rules that auditors can read; patient triage checklists.

Random Forest

Supervised | Tabular

HOW IT WORKS

Trains hundreds of decision trees, each on a random sample of rows and features, then lets them vote. Combining many trees cancels out individual mistakes.

REAL-WORLD USE

Credit card fraud detection; predicting which customers will cancel a subscription.

Gradient Boosting (XGBoost)

Supervised | Tabular

HOW IT WORKS

Builds small trees one after another, with each new tree focusing on the errors that the previous trees made.

REAL-WORLD USE

Credit scoring at banks; ranking search and ad results; a frequent winner of data science competitions on tabular data.

K-Nearest Neighbors (KNN)

Supervised | Tabular | Vision

HOW IT WORKS

Stores the training data and labels a new item by a majority vote among the K most similar examples, measured by distance.

REAL-WORLD USE

"Customers like you also bought" recommendations; recognizing handwritten digits.

Convolutional Neural Network (CNN)

Supervised | Vision | Generative AI

HOW IT WORKS

Slides small learned filters across an image to detect edges, then shapes, then whole objects, one layer at a time.

REAL-WORLD USE

Detecting tumors in medical scans; face unlock on phones; spotting defects on assembly lines.

UNSUPERVISED finds structure without labels

K-means Clustering

Unsupervised | Tabular | Vision

HOW IT WORKS

Places K center points, assigns every item to its nearest center, moves each center to the middle of its group, and repeats until the groups stop changing.

REAL-WORLD USE

Segmenting customers for targeted marketing; reducing the number of colors in an image.

Principal Component Analysis (PCA)

Unsupervised | Tabular | Vision

HOW IT WORKS

Finds the few directions that capture the most variation in the data and describes every item using only those, shrinking many features into a handful.

REAL-WORLD USE

Compressing hundreds of sensor readings before modeling; early face recognition systems (eigenfaces).

Transformer (BERT, GPT)

Unsupervised (self-supervised) | NLP | Generative AI | Vision

HOW IT WORKS

Uses attention so each word can weigh every other word in the text. It is pretrained on huge amounts of unlabeled text by predicting missing or next words, then often fine-tuned with labeled data.

REAL-WORLD USE

Chat assistants such as ChatGPT and Gemini; Google Translate; BERT improving Google Search results.

Generative Adversarial Network (GAN)

Unsupervised | Generative AI | Vision

HOW IT WORKS

Two networks compete. A generator creates fake samples, a discriminator tries to catch them, and both improve until the fakes look real.

REAL-WORLD USE

Photorealistic faces (StyleGAN); upscaling low-resolution photos; creating synthetic training data.

Diffusion Model

Unsupervised (self-supervised) | Generative AI | Vision

HOW IT WORKS

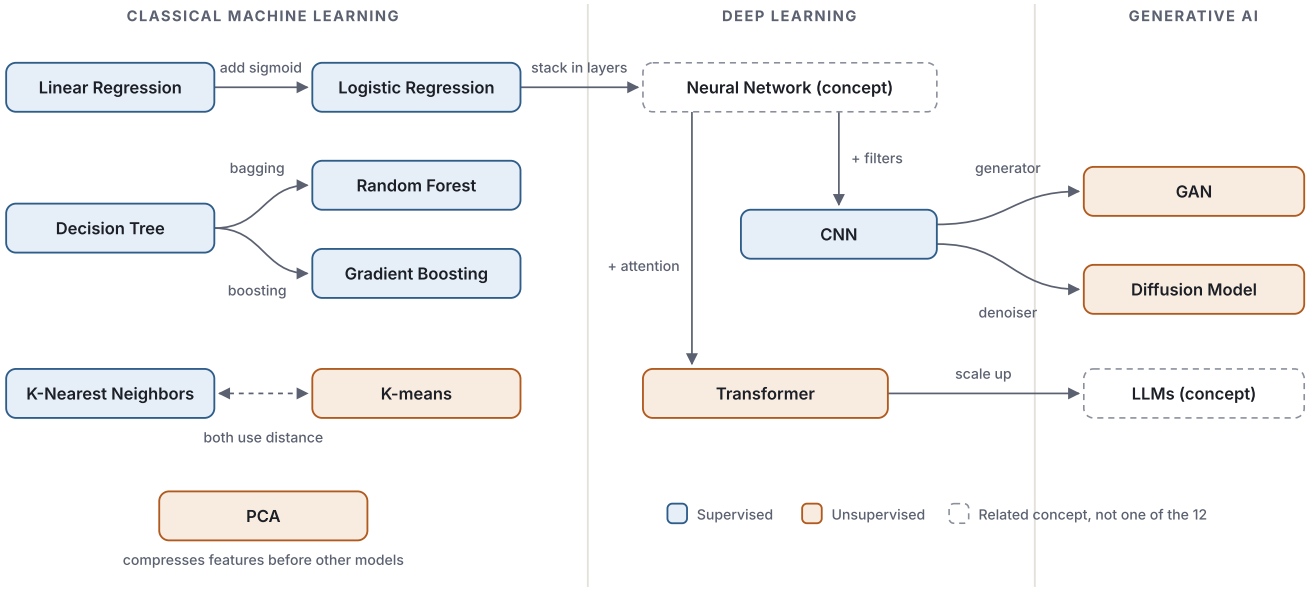
Learns to reverse a process that slowly adds noise to images. To generate, it starts from pure noise and removes it step by step, guided by a text prompt.

REAL-WORLD USE

Text-to-image tools such as Stable Diffusion, DALL·E 3 and Midjourney.

03 How the Algorithms Connect

These algorithms are not isolated. Many newer methods are built directly on older ones. The columns follow the familiar progression from machine learning to deep learning to generative AI.



One neuron is a logistic regression.
A neural network stacks many of these units in layers. Adding filters creates a CNN for images, and adding attention creates a Transformer for language.

Trees become teams.
Random Forest averages many independent trees (bagging), while Gradient Boosting trains trees in sequence so each one fixes earlier errors (boosting).

Similar names, different learning.
KNN and K-means both measure distance, but KNN needs labels to classify, while K-means groups unlabeled data on its own.

04 Choosing the Right Algorithm

The same two questions used to classify the algorithms in this framework can guide which one to use for a real problem.

Question 1: Do you have labeled examples?
Labeled means each example already has the correct answer, such as "fraud" or "not fraud."

YES → SUPERVISED

Question 2: What kind of data, and what do you need?

Table, predict a number	**Linear Regression**
Table, explainable yes or no	**Logistic Regression, Decision Tree**
Table, highest accuracy	**Random Forest, XGBoost**
Find similar examples	**KNN**
Images	**CNN**
Text classification	**Fine-tuned Transformer**

NO → UNSUPERVISED

Question 2: What is the goal?

Discover natural groups	**K-means**
Simplify many features	**PCA**
Generate new text	**Transformer (LLM)**
Generate realistic images	**GAN**
Generate images from a prompt	**Diffusion Model**

05 How the Algorithms Were Classified

Learning type
An algorithm is supervised if training requires labeled answers. It is unsupervised if it learns patterns from unlabeled data. Self-supervised models create their own training targets from raw data, so they are grouped with unsupervised learning.

Application domain
Domains are based on the type of input data the algorithm is mainly built for: rows and columns (tabular), images (vision), text (NLP), or creating new content (generative AI).

Primary versus secondary
A filled dot marks the domain where the algorithm is standard practice today. An open dot marks a real but less common use, such as K-means for compressing image colors.